

Week 5 Assignment: Tabular Q-Learning

A tested model-free agent with controlled exploration and oracle comparison

CMPSC 480 - Student Question Deck

Assessment overview

Train a tabular Q-learning agent and evaluate learning behavior against a planning oracle.

Coverage: Week 5: TD learning, exploration, terminal targets, and empirical evaluation

Points

10 total

Expected time

6-8 hours

Submission

Repository, learning curves,
and report

Learning objectives

- ▶ Separate environment dynamics from agent state.
- ▶ Implement epsilon-greedy action selection with deterministic evaluation.
- ▶ Apply the terminal-aware Q-learning update.
- ▶ Run reproducible training and evaluation phases.
- ▶ Compare a learned policy with value-iteration ground truth.

Requirements checklist

- ▶ Use Python 3.11 and NumPy only unless the prompt explicitly permits another package.
- ▶ Preserve the published interfaces and make all tie-breaking deterministic.
- ▶ State assumptions, validate inputs, and handle empty, terminal, and unreachable cases.
- ▶ Include focused tests whose names explain the defect each test would expose.
- ▶ Report measured evidence; do not replace experimental results with unsupported claims.
- ▶ Seed every source of randomness used in reported experiments.

Task 1 - Agent and exploration contract (2 points)

Implement a Q-table agent without embedding environment transition logic.

- a. Define observe, select_action, update, and evaluate-mode behavior. [1 points]
- b. Specify epsilon-greedy selection and a bounded decay schedule. [1 points]

Task 2 - Q-learning update (3 points)

Implement and validate the off-policy temporal-difference update.

- a. State the update and define the TD error. [1 points]
- b. Compute one update for $Q=2$, $r=1$, $\gamma=0.9$, max next $Q=4$, and $\alpha=0.5$. [1 points]
- c. Explain and test terminal handling. [1 points]

Task 3 - Training protocol (2 points)

Build a reproducible episodic training loop with safety limits.

- a. Describe reset, interaction, update, and episode termination order. [1 points]
- b. Log return, episode length, epsilon, and moving-average return. [1 points]

Task 4 - Evaluation and tests (3 points)

Measure policy quality and diagnose sensitivity to exploration and rewards.

- a. Evaluate the greedy policy without learning or exploration. [1 points]
- b. Compare greedy actions or returns with a value-iteration oracle. [1 points]
- c. Run one controlled hyperparameter ablation and cover sparse-reward and unseen-state cases. [1 points]

Edge cases and defensive programming

- ▶ No legal actions are returned.
- ▶ The transition is terminal.
- ▶ All action values tie.
- ▶ A state has never been visited.
- ▶ An episode reaches the safety step limit.

Submission instructions

- ▶ Submit source, tests, seeds, and hyperparameters.
- ▶ Include learning curves plus a compact oracle-comparison table.
- ▶ Document the exact command that reproduces the reported run.

Do not submit credentials, generated caches, or unapproved libraries.

Grading rubric

Category	Points	Required evidence
Agent and exploration contract	2	Agent API, epsilon schedule, and action-selection tests.
Q-learning update	3	Update function and hand-computed unit tests.
Training protocol	2	Training loop, seeds, and logged metrics.
Evaluation and tests	3	Oracle comparison, ablation, and edge-case tests.

Reflection prompt

In 150-200 words:

Explain one gap between a learned policy and the planning oracle. Distinguish modeling mismatch from sampling or optimization error.